Arc Welding Machine



Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Arc Welding Machine (Transfor mer); Arc Welding Machine (Transformer) NABL test certified; Welding Machine

What it is used for

An arc welding machine supplies electrical current to create an arc that melts and joins compatible metal parts. Correct current settings, secure work connections, and suitable protective equipment help produce sound welds while reducing electric shock, fire, and eye injury risks.

Main parts

- Power source or transformer supplies welding current.
- Current control adjusts the welding output.
- Electrode holder grips the consumable electrode.
- Welding electrode carries current and provides filler material where applicable.
- Work clamp connects the return cable to the workpiece.
- Welding cables carry current between the machine and work area.
- Power switch controls the machine supply.
- Cooling vents and protective casing support safe operation.

Before you start

1. Only trained personnel should perform arc welding under an approved workshop procedure.
2. Read the machine manual and confirm the process, electrode type, and current range for the material.
3. Inspect the machine, electrode holder, work clamp, cables, plug, and insulation for damage.
4. Ensure the workpiece is clean, securely positioned, and connected to the return clamp.
5. Remove flammable materials and check the area for fire hazards before welding.
6. Wear a suitable welding helmet with the correct shade, welding gloves, protective clothing, and safety footwear.

How to use

1. Position the machine on a stable, dry surface with adequate ventilation.
2. Connect the power supply and welding cables according to the manual.
3. Attach the work clamp securely to a clean metal area on the workpiece.
4. Set the welding current and polarity according to the electrode and material requirements.
5. Fit the correct electrode into the holder without touching exposed conductive parts unnecessarily.
6. Place suitable welding screens to protect nearby people from arc radiation and sparks.
7. Strike the arc using the approved technique and maintain a controlled arc length.

8. Move the electrode steadily along the joint while monitoring weld appearance and machine behaviour.
9. Stop welding if cables overheat, the machine behaves abnormally, or a fire hazard appears.
10. Switch off the machine after welding, allow hot parts to cool, and inspect the weld safely.

⚠ Safety

- Never weld with wet gloves, wet clothing, or in standing water.
- Do not touch the electrode or exposed conductive parts while the circuit is energised.
- Use suitable eye, face, hand, and body protection against arc radiation, sparks, and hot metal.
- Provide adequate ventilation and use appropriate fume extraction for the materials being welded.
- Keep a suitable fire extinguisher available and monitor the area for smouldering material after welding.
- Do not weld sealed containers or unknown materials without an approved assessment and preparation procedure.

Care & storage

- Switch off and isolate power before cleaning or inspecting the machine.
- Remove dust from vents and the exterior using an approved method.
- Inspect cables, holders, clamps, connectors, and insulation regularly for wear or damage.
- Arrange qualified servicing for overheating, abnormal output, damaged insulation, or repeated tripping.

Common mistakes to avoid

- Using an incorrect current setting or unsuitable electrode.
- Welding with a loose work clamp or damaged cable insulation.
- Working without a proper welding helmet or nearby fire precautions.
- Ignoring fumes, overheating, or damaged electrical components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.